

STATEMENT OF WORK

1. Objective/Requirements

This requirement is for the acquisition of the FY26 Total Materia Renewal for NASA Headquarters (HQ). The information listed below is required to complete this purchase.

2. Characteristics, Scope, and Specs

QTY	DESCRIPTION
1	Total Materia Basic Package + Extended Range + SmartComp + Material Console + Compliance; Corporate License, Annual Components: Total Materials, DataPLUS, Suppliers, Tracker, Extended Range, SmartComp, Material Console, Compliance

The Contractor shall provide access to the Materials Database to the NASA user community comprised of NASA personnel and on-site contractors residing at each of the 15 NASA Facilities, including ancillary facilities, as follows:

- Location 1: Ames Research Center (ARC).
- Location 2: Armstrong Flight Research Center (AFRC).
- Location 3: John H. Glenn Research Center (GRC).
- Location 4: Robert H. Goddard Space Flight Center (GSFC).
- Location 5: Jet Propulsion Laboratory (JPL).
- Location 6: Lyndon B. Johnson Space Center (JSC).
- Location 7: John F. Kennedy Space Center (KSC).
- Location 8: Langley Research Center (LaRC).
- Location 9: George C. Marshall Space Flight Center (MSFC).
- Location 10: NASA Headquarters (HQ).
- Location 11: John C. Stennis Space Center (SSC).
- Location 12: Michoud Assembly Facility (MAF).
- Location 13: White Sands Test Facility (WSTF)/White Sands Complex (WSC).
- Location 14: Wallops Flight Facility (WFF).
- Location 15: Independent Verification and Validation (IV&V) (NASA Katherine Johnson IV&V Facility” in Fairmont, West Virginia)

The Contractor shall provide direct, seamless, electronic access to the Materials Database from NTSS and for transfer of a user’s profile from NTSS to the Materials Database that includes the following data, at a minimum:

- User’s universally unique identifier (UUID) (acquired via the Security Assertion Markup Language (SAML) login).
- User’s first name (acquired via the SAML login).
- User’s last name (acquired via the SAML login).
- User’s email (acquired via the SAML login).

- User's Center (acquired via the SAML login).

The Contractor shall provide on-demand access via electronic authorization (SAML) to users who have been identified and authenticated in the NASA Account Management System (NAMS).

The Contractor shall provide a Web-based, user-friendly, robust architecture that is seamlessly integrated for effectively accessing materials data.

The Contractor shall provide search capability for users to find and export material information quickly and efficiently by:

- Material designation, searching by the alphanumeric or numerical material designation.
- Property values, mechanical and physical.
- Standard number.

The Contractor shall provide a one-stop shop for engineering materials information that includes built-in tools to help the user choose the most optimal material for the design, with sharing capabilities to limit repetition within an organization.

The Contractor shall provide multiple material groups and material properties, including metals (ferrous and non-ferrous), non-metals (polymers, ceramics, composites, fibers, cement, honeycombs, foams, wood), coatings, lubricants, adhesives, and substances.

The Contractor shall provide general material properties and information as follows:

- Material description, highlighting its applications, standard, and/or processes.
- Various temperature values and various conditions for each material designation.
- Mechanical properties.
- Physical properties.
- Chemical composition.
- Heat treatment diagrams and standard descriptions.
- Metallography diagrams, including microstructure diagrams in various grain sizes.
- Machinability and manufacturing processes information.
- Global equivalents list.

The Contractor shall provide advanced material properties for the following:

- Stress/strain diagrams.
- Formability data, including high-strain curves, forming limit curves, bendability data, and r-factors.
- Fatigue data, including total, plastic, and elastic curves; experimental points; and S-N curves.
- Fracture mechanics, including KIC values and Paris Law curves.
- Creep data, including creep rupture strength data tables and Larson-Miller parameter estimator.

The Contractor shall provide the following additional data:

- Welding, brazing, and adhesion information, including material weld properties, consumable property data, weld regime information, and additional joining information.
- Tribology information such as how two surfaces behave in motion.
- Corrosion information, including, but not limited to, corrosion resistance data, pitting resistance equivalent number (PREN), pitting, and chemical resistance.
- Weatherability information, including property values of materials in various environmental conditions.
- Ageing information, including time/temperature dependency information and diagrams and fluid degradation data.
- Irradiation, including exposure time versus radiation diagrams.
- Suppliers' information, including contact information.
- Compliance information, including compliance regulations for non-metals and substances.

The Contractor shall provide the capability to create material lists, bookmarks, and material reports with options to save, share information, and download material and compliance reports in portable document format (PDF).

The Contractor shall provide the capability to download and export material information to computer-aided engineering (CAE)/computer-aided design (CAD) solvers and Microsoft Excel.

The Contractor shall provide the following tools:

- Unknown metals identification tool, an intelligent metal identification system by chemical composition obtained from a spectrometer or other analytical sources through a patented algorithm.
- Compliance regulations assessments that instantly assesses materials' compliance using composition data.
- Data information tracking tool with the ability to monitor monthly updates, specifically the changes and additions to each material and standard revisions and replacements.
- Material comparison methods for multiple ways to compare specific material properties for several materials at once.
- Similar materials search engine for finding materials with similar property values such as mechanical and physical properties, material group, and chemical composition.

The Contractor shall provide the capability to create material lists, bookmarks, and material reports with options to save, share information, and download material and compliance reports in PDF.

The Contractor shall provide the capability to download and export material information to CAD/CAD solvers and Microsoft Excel.

Current database content and addition of new materials and datasets shall be updated at least monthly.

The Contractor shall provide a Materials Database via NTSS that complies with Section 508 of the Rehabilitation Act.

The Contractor shall have detailed quality control procedures in place to ensure requirements of the contract are met and the data in the Materials Database are current, accurate, and timely.

The Contractor shall provide advance notice to the Government if the Materials Database will be temporarily unavailable on the Website and of potential system performance impacts in advance of scheduled outages for maintenance.

The Contractor shall schedule planned downtime during off-duty hours if necessary for maintenance.

The Contractor shall notify the Government immediately if the Materials Database becomes inaccessible and provide reasons for the downtime.

The Contractor shall provide notification to the Government when the Materials Database becomes accessible following all downtime.

The Contractor shall provide technical customer support by personnel with detailed knowledge of the Materials Database to respond to Government user issues 24 hours a day/7 days a week that includes digitizing and curating material information based on user requests; programming, testing, and adding additional functionalities and tools to the database based on user requests; interpreting user requests; and providing support with the most challenging metals identification, properties search, and general standards information.

The Contractor shall provide feedback to the Government regarding types of issues reported by NASA users.

The Contractor shall provide unlimited Web-based training on use of the Materials Database and prior to release of new features and enhancements.

3. Place of Performance

E-delivery to NASA HQ
ATTN: Tracy Osborne

4. Period of Performance

05/21/26 – 05/31/27
Need by: 05/20/26